GUIDE INTERACTION DIARY FORM

We, the undersigned, **Shivam Maurya** (Seat No. **502511822**) and **Manoj Nadar** (Seat No. **502511827**), currently pursuing the **Third Year of the Bachelor of Science in Data Science (B.Sc. Information Technology) for the Academic Year 2026–2027** at **Sree Narayana Guru College of Commerce (Autonomous)**, hereby declare that we are undertaking our **Community Engagement Project** under the guidance of **Ms. Manali M. Kotkar**

We confirm that we have met our Internal Guide on the dates mentioned below for **project guidance, discussion, review, and progress monitoring**.

Sr.No	Date	Signature of the Internal Guide

Signature of the Students
(Shivam Maurya, Manoj Nadar)

Signature of the Internal Guide

PROFORMA FOR THE APPROVAL PROJECT PROPOSAL

(Note: All entries of the proforma of approval should be filled up with appropriate and complete information. Incomplete proforma of approval in any respect will be summarily rejected.)

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2. Title of the Project: NutriTrack – Nutrition Tracking System

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ABSTRACT

NutriTrack is an Android-based Nutrition Tracking System designed to help everyday users understand and manage their daily food, water and body-weight habits. The application combines a dashboard with calorie and water goals, dual progress rings, a streak badge and one-tap quick-add buttons with interactive height and weight sliders that calculate BMI and show the WHO healthy weight range. Three AI-powered features are provided through the Groq API using a Qwen model: a Food Macro Analyzer that estimates calories, protein, carbohydrates, fats and a health score from a food photograph, a Leftover Recipe Generator and a Face Analyzer for general wellness hints. The app is built in Android Studio with Kotlin and XML, uses a dark Material Design interface with bottom navigation and keeps user data in the local cache of the device. NutriTrack addresses the lack of everyday awareness about nutrition, hydration and BMI, reduces the effort of manual food logging and helps to cut household food waste. All AI results are estimates meant for guidance only and are not medical advice.

ACKNOWLEDGEMENT

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Date:

Place: Chembur

Completed by

Shivam Maurya
Manoj Nadar

DECLARATION

I hereby declare that the project entitled, "**NutriTrack – Nutrition Tracking System**" done at **Sree Narayana Guru College of Commerce (Autonomous)**, has not been in any case duplicated to submit to any other university for the award of any degree. To the best of my knowledge other than me, no one has submitted to any other university.

The project is done in fulfillment of the requirements for the award of a degree of **BACHELOR OF SCIENCE (INFORMATION TECHNOLOGY)** is to be submitted as a Third semester Community Engagement Project as part of our curriculum.

Shivam Maurya
Manoj Nadar

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1. INTRODUCTION

1.1 Background

Food is the foundation of health, yet most people have very little idea of what they actually eat and drink in a day. Urban lifestyles, long study and working hours, packaged food and irregular meal timings have made it difficult to follow a balanced diet. Lifestyle-related problems such as obesity, low energy, poor hydration and nutrient imbalance are now seen even among students and young adults. Understanding calories, protein, carbohydrates, fats and daily water intake is the first step towards healthier habits, but calculating these values manually is slow and boring, so most people never do it.

Smartphones have changed what is possible. A modern Android phone has a good camera, a fast processor and an always-available internet connection. Together with recent advances in artificial intelligence, especially multimodal models that can understand both images and text, a phone can now look at a plate of food and estimate its nutritional content within a few seconds. This project, **NutriTrack**, uses these possibilities to create a simple, friendly and practical Nutrition Tracking System for everyday users.

1.2 Community Problem

In the local community, especially among college students, working professionals, homemakers and elderly people, healthy eating is mostly based on guesswork. The common difficulties observed are:

- People do not know the approximate calories and macronutrients of the food they eat, especially Indian home-style dishes such as upma, poha, dal-rice or paratha.
- Water intake is rarely tracked, so many people remain mildly dehydrated without noticing it.
- Existing nutrition applications often need manual searching and typing of every food item, which people stop doing after a few days.
- Many applications are in complex layouts, filled with advertisements or locked behind paid plans.
- Leftover ingredients at home are often wasted because people do not know what can be cooked from them.
- People do not regularly check simple health indicators such as Body Mass Index (BMI) against a recognised standard.

1.3 Proposed Solution: NutriTrack

NutriTrack is an Android application developed in Android Studio using Kotlin for the application logic and XML for the user-interface layouts. It brings the most useful daily wellness functions into one app with a clean, dark-themed interface and a bottom navigation bar. The application provides:

- **Dashboard:** a daily goal view with a dual-ring progress chart (calories on the outer ring, water on the inner ring), a streak badge, quick-add buttons for water (+250 ml, +500 ml) and calories (+250 kcal, +500 kcal), a reset option, and quick-access tools.
- **Interactive Biometrics and BMI:** height and weight sliders that instantly calculate BMI, show the category (such as Normal) and display the WHO healthy weight range, with a button to save the biometrics to the profile.
- **Food Macro Analyzer:** the user takes a photo with the camera or selects one from the gallery; the AI identifies the dish and returns calories, protein, carbohydrates, fats and a health score out of 10.
- **Recipe Generator:** the user types leftover ingredients separated by commas, and the AI returns a recipe with a title, cooking time and step-by-step instructions.
- **Face Analyzer:** the user provides a face photo, and the AI gives a general skin-health observation, a fatigue index and simple care suggestions such as hydration and sun protection.
- **BMI Calculator:** a separate screen where height and weight can be typed and the BMI and category are shown with a clear tick or warning symbol.

1.4 Scope

- Awareness of balanced nutrition, hydration and healthy BMI among ordinary users.
- Daily tracking of water intake and calorie intake against personal goals.
- AI-based estimation of macronutrients from a food photograph.
- AI-based recipe suggestions from available ingredients to reduce food waste.
- General wellness observations from a face image (not a medical diagnosis).
- A lightweight Android application that works on common smartphones running modern Android versions.

1.5 Important Clarification

NutriTrack is a wellness-awareness and tracking prototype. The nutritional values, health scores, skin observations and fatigue index produced by the AI are **estimates** based on visual appearance and general knowledge. They are not laboratory measurements and they are not medical advice. Portion size, cooking oil, ingredients and preparation method can change the real values significantly. Users with medical conditions, allergies or special dietary needs must consult a qualified doctor or dietitian. The screenshots shown in this report use demonstration data.

1.6 Technology Stack

Table 1.1: Technology stack of NutriTrack

Layer	Technology used
IDE	Android Studio
Programming language	Kotlin (application logic, activities, fragments, API calls)
User interface	XML layouts with Material Design components, bottom navigation, cards, sliders and buttons
AI engine	Groq API serving a Qwen large language model (model name set in the code) for image/text understanding and text generation
Input	Device camera and phone gallery for images; text fields for ingredients, height and weight
Local data	Local cache on the device for user login details, daily trackers, goals and saved biometrics
Backend	No Firebase and no separate server; the app calls the Groq API directly
Networking	Internet connection (mobile data or Wi-Fi) for AI requests
Target platform	Android smartphones

1.7 Social Impact

By making nutrition information easy to obtain, NutriTrack encourages people to think before they eat, to drink water regularly and to monitor their body weight in a healthy range. The recipe generator reduces household food waste, which is also a social and environmental concern. The application is free of complicated medical terms and uses simple numbers and colours, so that even first-time users can understand their daily progress.

1.8 Need for the Project

The need for the project arises from the following observations:

- Manual calorie counting needs food-composition tables and calculations, which most people find difficult.
- A photograph is the easiest possible input; a person only has to point the camera at the food.
- Hydration and calorie goals are most effective when they are visible on one screen every day.
- A streak counter and quick-add buttons make tracking a habit instead of a chore.
- Indian users need support for local dishes, which many international applications do not recognise well.

1.9 Problem Statement

People need a simple, quick and low-cost mobile tool that can estimate the nutritional value of their food from a photograph, track daily water and calorie intake, calculate BMI, suggest recipes from leftover ingredients and give basic wellness observations, all in one application, while clearly reminding them that the results are estimates and not medical advice.

1.10 Organization of the Report

The report is divided into eleven chapters: introduction (Chapter 1), objectives (2), literature review (3), project content and case studies (4), research methodology (5), survey analysis (6), field visit documentation (7), design, implementation and results (8), findings and discussion (9), conclusion and recommendations (10) and references (11).

2. OBJECTIVES OF THE PROJECT

1. To understand the nutrition, hydration and healthy-living needs of people in the local community.
2. To explain the importance of calories, protein, carbohydrates, fats and water in everyday language.
3. To build a dashboard that shows the daily calorie and water goals together in one view.
4. To provide quick-add buttons so that water and calorie intake can be logged within a second.
5. To provide interactive height and weight sliders that calculate BMI and show the WHO healthy weight range.
6. To develop a Food Macro Analyzer that estimates calories, protein, carbohydrates, fats and a health score from a food photograph.
7. To develop a Leftover Recipe Generator that suggests a recipe from the ingredients available at home.
8. To develop a Face Analyzer that gives general skin-health and fatigue observations with care tips.
9. To provide a separate BMI Calculator for quick manual calculation.
10. To design a dark-themed, easy-to-read Android interface using XML and Kotlin.
11. To encourage responsible use of AI estimates and to remind users that they are not a substitute for medical advice.
12. To document the community need and relate the software prototype to real users.
13. To apply Android Studio, Kotlin, XML and AI API integration to a practical community problem.

2.1 Expected Outcomes

The expected outcome is a functional Android application that places the most common daily wellness tasks in one interface. A user should be able to log water and calories within a second, find the nutritional values of a meal by taking a photograph, check BMI against a recognised range and obtain a recipe from leftover ingredients.

2.2 Explanation of the Objectives

The thirteen objectives can be grouped into three categories: awareness objectives, functional objectives and learning objectives.

Awareness objectives (Objectives 1, 2, 11 and 12). These focus on understanding the needs of the community, explaining nutrition in simple language and encouraging responsible use of the app.

Functional objectives (Objectives 3 to 10). These describe what the NutriTrack prototype actually does. The dashboard shows the progress rings and trackers, the biometrics card calculates BMI, the Food Macro screen analyses images, the Recipe Gen screen creates recipes, the Face Scan screen gives wellness observations and the BMI Calculator performs a quick calculation. All screens are connected through a bottom navigation bar.

Learning objective (Objective 13). The project gives practical experience in Android development with Kotlin and XML, camera and gallery access, REST-style API integration with a generative AI model, handling of asynchronous responses and design of a user-friendly interface.

3.2 Hydration

Water forms a large part of the human body and is needed for temperature control, digestion, transport of nutrients and removal of waste. Mild dehydration can cause tiredness, headache and difficulty in concentration, and it can also make the skin look dull. General guidance suggests that an adult needs roughly two to three litres of total fluid per day, depending on climate and activity. A visible water goal with one-tap logging, as provided by NutriTrack, makes it easier to build the habit of drinking water regularly.

3.3 Body Mass Index (BMI)

Body Mass Index is a simple screening measure calculated by dividing weight in kilograms by the square of height in metres:

BMI = Weight (kg) / [Height (m) x Height (m)]

The World Health Organization classifies adult BMI into categories, and the range of 18.5 to 24.9 is considered the normal or healthy range. From this range, a healthy weight band can be calculated for any height. For example, a person with a height of 175 cm has a healthy weight band of approximately 56.7 kg to 76.3 kg, which is the same band shown by the NutriTrack biometrics card. BMI is only a screening tool: it does not directly measure body fat, and it may not be accurate for athletes, older adults or people with unusual body composition.

Table 3.2: WHO adult BMI categories

BMI value	Category
Below 18.5	Underweight
18.5 to 24.9	Normal
25.0 to 29.9	Overweight
30.0 and above	Obese

3.4 Mobile Health and Nutrition Applications

Mobile health (mHealth) applications have become popular for fitness tracking, diet logging, step counting and reminders. Studies and surveys on such applications commonly report that regular self-monitoring helps people become aware of their habits, and that simple design, quick data entry and visible progress make people more likely to continue using the application.

A frequently reported weakness of traditional calorie-counting applications is the effort needed to enter every meal manually. Users have to search a database, select a food item and estimate the quantity. This extra effort is one reason why many users stop logging after a short time.

3.5 Artificial Intelligence in Food Recognition

Computer vision has long been used to classify images of food. Earlier systems relied on convolutional neural networks trained on fixed food datasets and could recognise only the classes present in training data. Newer multimodal generative models can accept an image together with a text instruction, understand what is visible, and respond in natural language or in a structured format such as JSON.

Such models are not perfect. They estimate from the visual appearance, so hidden ingredients such as oil, sugar or butter and the exact portion size can lead to errors. For this reason, results from AI food analysis should be treated as approximate guidance and not as exact measurements.

3.6 Generative AI for Recipe Suggestions

Large language models can combine a list of ingredients with their knowledge of cooking methods and create a complete recipe with steps and an approximate cooking time. For households, this helps in using leftover ingredients, saving money and reducing food waste. In NutriTrack, the user enters ingredients such as rice, tomatoes and egg, and the model returns a recipe such as Tomato Egg Fried Rice with a cooking time and ordered steps.

3.7 Face Image Analysis for General Wellness

Skin appearance can reflect lifestyle factors such as hydration, sleep, sun exposure and stress. Cosmetic and wellness applications analyse a face photograph to comment on visible features such as dryness, texture, pores or dullness. These observations are cosmetic and general. They are not a medical examination and cannot diagnose any skin condition or disease. In NutriTrack the Face Analyzer is included as a wellness-awareness feature that gives simple suggestions such as using a moisturiser, applying sunscreen and drinking enough water, and it always has to be understood as general guidance.

3.8 Privacy and Responsible Use of Images

- A food or face image is personal data; it should be sent for analysis only with the knowledge of the user.
- Images of other people should not be analysed without their permission.
- API keys and credentials must never be hard-coded in a published application.
- The application should clearly tell the user that results are estimates and not medical advice.
- Stored data such as height, weight and goals should remain on the device unless the user chooses otherwise.

3.9 Existing Applications and the Project Gap

Popular calorie and fitness applications provide large food databases, barcode scanners and detailed analytics, but many of them require registration, subscriptions or heavy manual entry. Other applications focus on only one function, such as a water reminder or a BMI calculator, so users have to install several applications. NutriTrack aims to fill this gap for the community by combining hydration tracking, calorie tracking, BMI, photo-based macro analysis, recipe suggestion and face-based wellness hints in one lightweight application designed around everyday Indian foods and ingredients.

3.10 Role of Android, Kotlin and XML

- **Android Studio** is the official IDE for Android development and provides the editor, emulator, layout designer and debugging tools.
- **Kotlin** is the officially recommended language for Android. It is concise, null-safe and supports coroutines for background work such as network calls.
- **XML** is used to describe the screen layouts, including cards, sliders, buttons, text fields, image views and the bottom navigation bar.
- **Material Design components** give consistent buttons, cards and navigation behaviour.
- **Activity Result APIs** are used for selecting images from the gallery and capturing photographs with the camera.
- **Groq API** provides fast hosted inference of the Qwen model used for image and text understanding and for text generation.

3.11 Summary

The literature review shows that regular self-monitoring, simple interfaces and low-effort data entry are the main factors in the success of health applications. AI-based image understanding can remove the effort of manual food entry, but its results are estimates.

4. PROJECT CONTENT AND CASE STUDIES

4.1 Main Modules

Table 4.1: Main modules of NutriTrack

Module	Description
Dashboard	Greets the user and shows the daily goal (water in ml and energy in kcal), a dual-ring energy-goal chart, a streak badge, hydration and nutrition cards with quick-add buttons, a Reset button, the biometrics and BMI card and Quick Access Tools.
Interactive Biometrics and BMI	Height and weight sliders update BMI live, show the category and the WHO healthy weight target, and a Save button stores the values to the profile with a confirmation message.
Recipe Gen	Accepts comma-separated ingredients and generates a recipe with title, cooking time and numbered steps.
Face Scan (Face Analyzer)	Accepts a face image from the camera or gallery and returns a skin-health observation, a fatigue index and care suggestions.
Food Macro (Food Macro Analyzer)	Accepts a food image and returns the dish name, calories, protein, carbohydrates, fats and a health score out of 10.
BMI Calculator	A simple screen with Height (cm) and Weight (kg) fields and a Calculate BMI button that displays the BMI value and category.
Bottom Navigation	Four tabs - Dashboard, Recipe Gen, Face Scan and Food Macro - for fast movement between the main screens.

4.2 Detailed Description of the Modules

4.2.1 Dashboard Module

The dashboard is the home screen. At the top it shows a greeting with the name of the user and the daily goals, for example 2500 ml of water and 2400 kcal of energy. A circular chart in the middle uses two rings: the outer ring (red) represents calorie progress and the inner ring (green) represents water progress. The percentage of the energy goal is displayed at the centre. A flame-style streak badge shows the number of consecutive days the user has used the tracker. Below the chart, a Hydration card and a Nutrition card show the current value against the goal and provide buttons for adding 250 ml or 500 ml of water and 250 kcal or 500 kcal of energy. A Reset button clears the trackers for a new day and shows a short message that the trackers were reset to 0.

4.2.2 Interactive Biometrics and BMI Module

This card on the dashboard contains two sliders: one for height in centimetres and one for weight in kilograms. When the user moves a slider, the BMI is recalculated at once and displayed in large text together with a coloured label such as Normal. A line below shows the WHO healthy weight target for the selected height. The button SAVE BIOMETRICS TO PROFILE stores the values and shows a confirmation message such as the saved height, weight and BMI.

4.2.3 Food Macro Analyzer Module

On this screen the user sees a large image area with two buttons, Camera and Gallery. After an image is chosen, it is displayed in the image area, converted to a suitable size and sent with an instruction to the AI model. The response is shown in a result card containing the name of the dish, calories, protein, carbohydrates, fats and a health score out of 10. In the sample screenshot, a plate of wheat rava upma is identified and shown with 320 kcal, 10 g protein, 55 g carbohydrates, 8 g fats and a health score of 7 out of 10.

4.2.4 Recipe Generator Module

The user types the available ingredients in a text field labelled Ingredients, comma separated, and taps Generate Recipe. The application builds a prompt from the ingredients and sends it to the AI model. The response is displayed as a recipe title, cooking time and numbered steps. In the sample screenshot, rice, tomatoes and egg produce Tomato Egg Fried Rice with a cooking time of 20 minutes.

4.2.5 Face Analyzer Module

This screen follows the same Camera and Gallery pattern. After the image is analysed, a result panel shows a skin-health observation, a fatigue index (for example Moderate) and a short list of suggestions such as using a hydrating serum, applying a broad-spectrum sunscreen daily and using a gentle exfoliant twice a week. The result is general wellness guidance only.

4.2.6 BMI Calculator Module

The BMI Calculator has two input fields and a Calculate BMI button. When the user enters, for example, 170 cm and 65 kg, the screen shows BMI 22.5 with a tick symbol and the label Normal.

4.3 Input, Processing and Output of Each Module

Table 4.2: Input, processing and output

Module	Input	Processing	Output
Dashboard trackers	Taps on +250/+500 buttons	Adds to the stored water or calorie value and updates the rings	Updated values, percentage and rings
Biometrics and BMI	Slider positions	BMI = weight / height squared; category and WHO range computed	BMI value, category label, healthy weight range
Food Macro	Food image (camera/gallery)	Image sent with prompt to the AI model; response parsed	Dish name, kcal, protein, carbs, fats, health score
Recipe Gen	Ingredient text	Prompt created and sent to the AI model; response parsed	Recipe title, cooking time, steps
Face Scan	Face image	Image sent with prompt to the AI model; response parsed	Skin observation, fatigue index, tips
BMI Calculator	Height and weight	BMI formula and category check	BMI value and category

4.4 Working of NutriTrack

1. The user opens the application and lands on the Dashboard, where the daily goals and the current progress are visible.
2. During the day the user taps the quick-add buttons after drinking water or eating a meal to update the rings.
3. To find the nutritional values of a meal, the user opens Food Macro, takes a photograph or picks one from the gallery and reads the result card.
4. If the user wishes to log the calories, the value can be added to the nutrition tracker using the quick-add buttons.
5. When unused ingredients are available, the user opens Recipe Gen, types the ingredients and reads the generated recipe.
6. The user can open Face Scan for general wellness hints and care suggestions.
7. Height and weight are updated on the Dashboard biometrics card or the BMI Calculator, and the biometrics can be saved to the profile.
8. At the start of a new day the Reset button clears the trackers while the streak shows the continuity of use.

5. RESEARCH METHODOLOGY

5.1 Approach

The project follows a practical development approach: identify a community need, define requirements, design the screens, implement the modules in Android Studio, test the main flows on a device and document the results with screenshots. An iterative method was used, in which each module was built, run on a phone or emulator, corrected and then combined with the others through the bottom navigation bar.

5.2 Requirement Collection

- Identify the common difficulties in daily nutrition and hydration tracking.
- List the information the user needs on one screen: goals, progress, BMI and quick actions.
- Decide the AI-based features: food analysis, recipe generation and face analysis.
- Design simple navigation using four bottom tabs.
- Prepare prompts that return the required fields in a predictable format.
- Use screenshots from the working application to review each screen.

5.3 System Requirements

Table 5.1: Development and user requirements

Category	**Requirement**
Development computer	Laptop or desktop capable of running Android Studio (recommended 8 GB RAM or more)
IDE	Android Studio with Android SDK and an emulator or a physical device
Languages	Kotlin and XML
Libraries	AndroidX, Material Components, an HTTP client and JSON parsing for calling the Groq API, coroutines
Test device	Android smartphone with camera, gallery and internet access
Permissions	Camera and internet; access to images selected through the gallery
Account	Groq API key obtained from the Groq console, kept outside the source code in production

5.7 Testing Method

Testing verified that each screen opens from the bottom navigation, that the quick-add buttons update the values and rings, that BMI is calculated correctly for sample values, that images can be selected from the camera and gallery, that the AI result cards display the returned fields and that the recipe screen returns a readable recipe. A sample manual calculation was used to confirm the BMI formula: for 170 cm and 65 kg, BMI = 65 / (1.70 x 1.70) = 22.5, which matches the output shown by the application.

5.8 Feasibility Study

Technical feasibility. Android Studio, Kotlin, XML and the Groq API are well documented and widely used. The application runs on ordinary smartphones, and the heavy AI processing happens on the server, so the phone does not need special hardware.

Economic feasibility. Android Studio is free, and the AI service offers a free tier suitable for development and demonstration. The only cost is a computer, a phone and an internet connection.

Operational feasibility. The interface uses large text, clear labels and a few buttons per screen. A user who can use a camera and a messaging application can learn NutriTrack within minutes.

5.9 Functional Requirements

1. The system shall display a dashboard with daily goals and progress for water and calories.
2. The system shall allow the user to add 250 ml or 500 ml of water and 250 kcal or 500 kcal of energy.
3. The system shall allow the trackers to be reset.
4. The system shall calculate BMI from height and weight and display the category and healthy weight range.
5. The system shall allow the biometrics to be saved to the profile.
6. The system shall accept a food image from the camera or gallery and display the estimated nutrition values and health score.
7. The system shall accept a list of ingredients and display a recipe.
8. The system shall accept a face image and display a general wellness observation, fatigue index and suggestions.
9. The system shall provide a BMI Calculator screen.

5.10 Non-Functional Requirements

Table 5.2: Non-functional requirements

Requirement	Description
Usability	Simple navigation, large readable text and high-contrast dark theme.
Performance	Local actions respond instantly; AI results appear within a few seconds depending on the network.
Reliability	The application should show a clear message when the network or AI service is unavailable instead of closing unexpectedly.
Security	API keys should not be exposed in the published code; images should be sent only after user action.
Privacy	Personal data such as height and weight should stay on the device.
Portability	The application should run on common Android versions and screen sizes.
Maintainability	Separation of layouts (XML), logic (Kotlin) and API calls makes changes easier.

5.11 Project Phases

Table 5.3: Phases of the project

Phase	Activity	Result
Planning	Identify the community problem and decide the modules	Module list and screen plan
Design	Prepare layouts, colours and diagrams	XML layouts and system diagrams
Development	Implement screens, trackers, BMI and AI integration	Working NutriTrack application
Testing	Check navigation, calculations, image input and AI output	Screenshots and testing checklist
Community study	Prepare the survey and visit documentation format	Questionnaire and visit format
Report writing	Compile screenshots, findings and recommendations	Final project report

Figure 6.1: Number of questions in each theme of the survey questionnaire

6.3 Analysis Method

After actual responses are collected, count the responses for each option and calculate the percentage using the formula:

Percentage = (Number of responses for an option / Total valid responses) x 100

The counts can be shown in tables, bar charts and pie charts, and the description should mention only what the collected data supports.

6.4 Interpretation Framework

- **Tracking habits:** compare how many people track water and calories with those who do not.
- **Awareness of BMI:** identify how many respondents know their BMI category.
- **Difficulties:** find the most common difficulty and check whether photograph-based input addresses it.
- **Feature preference:** identify the most requested module of NutriTrack.
- **Privacy concerns:** summarise the level of concern about sharing images and plan suitable guidance.

6.5 Questionnaire Response Sheet

Table 6.1: Format for recording survey responses

Q. No.	Topic of the question	Option / Answer	Count	%
1	Age group	18-25	3	100%
2	Tracks water intake	No	2	66.7%
3	Tracks calories	No	3	100%
4	Knows own BMI	No	2	66.7%
5	Method of finding meal calories	Guess	2	66.7%
6	Used a nutrition application before	No	2	66.7%
7	Biggest difficulty in tracking	Typing every item	2	66.7%
8	Interest in photo-based analysis	Yes	3	100%
9	Interest in recipe suggestions	Yes	2	66.7%
10	Most useful feature	Food scan	2	66.7%
11	Concern about sharing images	No	2	66.7%

The entries above are based on the three participants met during the field visit. As the sample is very small, the percentages are indications only.

6.6 Guidelines for Conducting the Survey

- Explain the purpose of the survey to every respondent before asking the questions.
- Take consent and do not collect names, phone numbers or medical details.
- Record the date, place and total number of respondents.
- Present only the results that were actually collected.
- Ask each question in simple language and, where needed, explain it in Hindi or Marathi so that the respondent understands it clearly.
- Read the options neutrally and do not influence or hint at the answer.
- Allow every respondent to skip any question that he or she does not want to answer.
- Do not photograph or analyse the face or food of a respondent without clear permission.
- Keep the filled response sheets safe and use them only for this project.

6.7 Limitation

Until actual responses are added, no statistical conclusion can be drawn about the wider community.

The following limitations should also be kept in mind when the survey is conducted and its results are written:

- The number of respondents is expected to be small and may consist mainly of college students, so the results may not represent other groups such as homemakers, working professionals or elderly people.
- The answers are self-reported and may not match the actual eating, drinking and exercise habits of the respondents.
- The survey is a snapshot of one visit and one day; habits and opinions may change over time.
- Opinions about photo-based analysis depend on how clearly the demonstration was understood and on the sample meal that was shown.
- The percentages calculated from a small sample should be described as indications only and not as proof.

7. FIELD VISIT DOCUMENTATION

7.1 Visit Details

This chapter records the community field visit connected with the project. The visit details below are taken from the geotagged photographs captured during the visit.

Table 7.1: Field visit record

Item	Details
Date of visit	Monday, 26/10/2026
Time	12:08 PM to 12:10 PM (GMT +05:30)
Place / Community visited	Pl Lokhande Marg, Chedda Nagar, Chembur West, Mumbai, Maharashtra 400089
Type of people met	College students
Number of people interacted with	3 (as seen in the geotagged photographs)
Persons from the college present	Students of Sree Narayana Guru College of Commerce (college ID cards visible)
GPS location (if photographs are geotagged)	Lat 19.061918, Long 72.909027 (GPS Map Camera)

7.2 Purpose of the Visit

The purpose of the visit is to connect the software project with real users: to understand how they currently manage food, water and body-weight monitoring, to demonstrate the NutriTrack application on a phone and to gather their opinion about its usefulness.

7.3 Suggested Activities During the Visit

- Explain the importance of balanced diet, hydration and BMI in simple language.
- Demonstrate the Dashboard trackers and show how water and calories are added.
- Show the Food Macro Analyzer with a sample meal and explain that the values are estimates.
- Show the Recipe Generator using ingredients available at the place.
- Demonstrate the BMI Calculator for volunteers who agree to check their BMI.
- Collect survey responses using the questionnaire in Chapter 6 with consent.

7.4 Field Observation Checklist

Table 7.2: Observation sheet

Observation	Yes / No	Remarks
Participants use a smartphone	Yes	All 3 used smartphones
Participants track water or calories at present	No	Only 1 of 3 tracks water
Participants know their BMI	No	2 of 3 did not know BMI
Participants found photo-based analysis easy	Yes	All 3 found it easy to use
Participants showed interest in recipe suggestions	Yes	2 of 3 were interested
Language or interface difficulty noticed	No	App understood without help
Concern about sharing images raised	Yes	1 of 3 asked about storage

Table 7.4: GPS log of the field-visit photographs

Figure	Date and time	Latitude	Longitude
7.1	26/10/2026, 12:08 PM	19.061921	72.909034
7.2	26/10/2026, 12:09 PM	19.061918	72.909027
7.3	26/10/2026, 12:09 PM	19.061918	72.909025
7.4	26/10/2026, 12:10 PM	19.061909	72.909026

7.6 Link Between the Field Visit and NutriTrack

Table 7.3: Expected link between observations and features

Possible observation	Related NutriTrack feature
People do not know the calories of what they eat	Food Macro Analyzer with calories, protein, carbs, fats and health score
People forget to drink water	Hydration tracker with +250 ml and +500 ml buttons
People are unaware of their BMI	Interactive biometrics card and BMI Calculator
Leftover food is wasted at home	Recipe Generator from available ingredients
Tracking is boring or takes too much time	Quick-add buttons, progress rings and streak badge

7.7 Learning from the Visit

- A short demonstration is more effective than a long explanation.
- The application must clearly say that its results are estimates and not medical advice.
- Users should be reminded to consult a doctor or dietitian for medical problems.

7.8 Ethical and Privacy Considerations

- Take permission before photographing people and before analysing anyone's face or food image.
- Write only the observations that were actually seen or recorded.

8.7.1 BMI Calculation and Healthy Weight Range

```
fun calculateBmi(heightCm: Int, weightKg: Double): Double {
    val h = heightCm / 100.0
    return weightKg / (h * h)
}
fun bmiCategory(bmi: Double): String = when {
    bmi < 18.5 -> "Underweight"
    bmi < 25.0 -> "Normal"
    bmi < 30.0 -> "Overweight"
    else -> "Obese"
}
fun healthyRange(heightCm: Int): Pair<Double, Double> {
    val h = heightCm / 100.0
    return Pair(18.5 * h * h, 24.9 * h * h) // 175 cm -> 56.7 to 76.3 kg
}
```

8.7.2 Quick-Add Tracker Update

```
private fun addWater(ml: Int) {
    waterTotal += ml
    updateRings()
}
private fun addCalories(kcal: Int) {
    calorieTotal += kcal
    updateRings()
}
private fun updateRings() {
    val calPercent = (calorieTotal * 100 / calorieGoal).coerceAtMost(100)
    val waterPercent = (waterTotal * 100 / waterGoal).coerceAtMost(100)
    outerRing.progress = calPercent
    innerRing.progress = waterPercent
    energyText.text = "$calPercent%"
}
```

8.7.3 Sending an Image to the AI Model (Groq API)

```
private fun analyseFood(base64Image: String) {
    lifecycleScope.launch(Dispatchers.IO) {
        try {
            val prompt = "Identify this dish. Reply with: Name, Calories, " +
                "Protein(g), Carbs(g), Fats(g), HealthScore(1-10)."
            val json = JSONObject()
                .put("model", MODEL_NAME) // Qwen model name
                .put("messages", JSONArray().put(JSONObject()
                    .put("role", "user")
                    .put("content", JSONArray()
                        .put(JSONObject().put("type", "text").put("text", prompt))
                        .put(JSONObject().put("type", "image_url").put("image_url",
                            JSONObject().put("url", "$base64Image"))))))
            val request = Request.Builder()
                .url("https://api.groq.com/openai/v1/chat/completions")
                .addHeader("Authorization", "Bearer $API_KEY")
                .post(json.toString().toRequestBody("application/json".toMediaType()))
                .build()
            val reply = client.newCall(request).execute().body?.string()
            withContext(Dispatchers.Main) { showResult(parseReply(reply)) }
        } catch (e: Exception) {
            withContext(Dispatchers.Main) { showError("Analysis failed. Check internet.") }
        }
    }
}
```

8.10 Testing Checklist

Table 8.2: Testing checklist

Test case	Expected result	Status
Bottom navigation	Each tab opens the correct screen	Verified in screenshots
Dashboard display	Goal, rings, streak and cards are visible	Verified in screenshots
Quick-add buttons	Totals and rings update	Verified (1000 mL / 1000 kcal shown)
Reset	Trackers return to 0 with a message	Verified in screenshot
BMI sliders	175 cm and 75 kg give BMI 24.5, Normal	Verified in screenshot
Save biometrics	Confirmation message with values	Verified in screenshot
BMI Calculator	170 cm and 65 kg give BMI 22.5, Normal	Verified in screenshot
Food Macro result	Dish name, kcal, macros and health score shown	Verified in screenshot
Recipe generation	Recipe with title, time and steps shown	Verified in screenshot
Face analysis	Observation, fatigue index and tips shown	Verified in screenshot

Note: Screenshots confirm visible behaviour. Behaviour such as error handling, rotation, different screen sizes and offline conditions should be tested separately.

8.11 Extended Test Scenarios

Table 8.3: Suggested additional tests (not claimed as completed)

Scenario	Expected result	Status
Open Food Macro and cancel image selection	No crash; screen stays unchanged	To be tested
Analyse an image with no food	Sensible message instead of wrong values	To be tested
Use the AI features without internet	Clear error message	To be tested
Leave ingredients empty and tap Generate Recipe	Asks the user to enter ingredients	To be tested
Enter zero or empty height or weight in BMI Calculator	Input rejected with a message	To be tested
Close and reopen the application	Saved biometrics and goals still available	To be tested
Rotate the phone or use a small screen	Layout remains readable	To be tested
Add calories above the goal	Percentage handled without breaking the ring	To be tested

8.14 User Guide

1. Install and open NutriTrack; the Dashboard appears first.
2. Check the daily goals shown below your name.
3. Tap +250 ml or +500 ml after drinking water, and +250 kcal or +500 kcal after eating.
4. Move the height and weight sliders to see your BMI and healthy weight range, and tap Save Biometrics to Profile.
5. Open Food Macro, tap Camera or Gallery, choose a food photo and read the result card.
6. Open Recipe Gen, type ingredients separated by commas and tap Generate Recipe.
7. Open Face Scan, choose a clear face photo and read the wellness hints.
8. Use Reset on the Dashboard to start a new day.

9.2 Discussion

NutriTrack shows how a student-level Android project can address a practical need by combining familiar mobile features (camera, gallery, sliders, buttons) with modern AI services. The most important design decision was to reduce effort: the user does not search a food database but simply takes a photograph, and the user does not remember a goal but simply looks at a ring.

At the same time, the project highlights the need for responsible communication. A health score or a skin observation can easily be mistaken for professional advice. For this reason the application and this report both emphasise that the results are guidance based on visual appearance. Future versions should show a short disclaimer on each AI result screen and encourage users to consult a doctor or dietitian when needed.

9.3 Benefits to the Community

- Promotes awareness of balanced diet, hydration and healthy BMI range.
- Helps beginners understand calories and macronutrients without technical knowledge.
- Encourages daily habits through goals, streaks and quick logging.

9.4 Comparison: Manual Tracking and NutriTrack

Table 9.1: Comparison of manual tracking and NutriTrack

Aspect	Manual / notebook method	NutriTrack prototype
Finding calories of a meal	Search tables or the internet and calculate	Take a photograph and read the estimate
Water tracking	Memory or marks on a bottle	One-tap +250 ml / +500 ml buttons and ring
Daily summary	Must be added by hand	Percentage and rings shown instantly
BMI check	Calculator and chart needed	Sliders with live BMI, category and WHO range
Meal ideas from leftovers	Search recipes manually	Type ingredients and get a recipe
Motivation	Depends on the person	Streak badge and visible goals
Accuracy	Depends on the person's knowledge	Estimated; depends on image and portion

9.5 Limitations

The report does not include a completed survey dataset, a confirmed number of participants or measured before-and-after results; therefore no quantified improvement in eating habits is claimed. The application is a prototype that depends on an external AI service and an internet connection. Its nutritional values are not laboratory measurements, and its face-based observations are not medical findings. The stored data is local to the device.

9.6 Future Improvements

- Add a food log history with daily, weekly and monthly charts.
- Allow the user to edit portion size and automatically add the analysed calories to the nutrition tracker.
- Add user-defined goals based on age, sex, height, weight and activity level.
- Add optional cloud backup and account sync so that data is not lost when the device is changed.
- Add Hindi, Marathi and other regional languages.
- Support Indian food databases and barcode scanning for packaged food.
- Add reminders for drinking water and for meal times.
- Show a short disclaimer on every AI result screen and use a consistent score scale.
- Add offline calculations for BMI and trackers with clear handling of network errors.
- Protect the API key using a secure backend instead of storing it in the application.

9.7 Learning Outcomes

- Designing user interfaces in XML with Material Design components.
- Writing application logic in Kotlin, including calculations and event handling.
- Using camera and gallery access in an Android application.
- Integrating a generative AI API and designing prompts for structured output.
- Using coroutines for background network calls.

10. CONCLUSION AND RECOMMENDATIONS

10.1 Conclusion

NutriTrack - Nutrition Tracking System demonstrates how a simple Android application, built with Kotlin and XML in Android Studio, can help people understand and manage their daily nutrition and hydration. The application combines a dashboard with calorie and water goals, a streak badge, quick-add buttons, interactive BMI sliders, a separate BMI calculator, an AI-based Food Macro Analyzer, a Leftover Recipe Generator and a Face Analyzer for general wellness hints.

The project shows that photograph-based input can remove much of the effort of manual food logging, and that clear visual goals can make daily tracking easier. It also shows that AI results must be presented honestly: they are estimates and not medical advice. With improvements in accuracy, privacy, language support and data history, NutriTrack can grow into a practical tool for students, families and the wider community.

10.2 Recommendations

1. Use NutriTrack as an awareness and guidance tool, and consult a doctor or dietitian for medical conditions or special diets.
2. Treat AI-estimated calories and health scores as approximate, especially when portion sizes are large or ingredients are hidden.
3. Take food photographs in good light with the dish clearly visible for better results.
4. Set realistic water and calorie goals according to age, activity and professional advice.
5. Do not analyse or share the images of other people without their permission.
6. Protect the API key and avoid sharing it in screenshots or public repositories.
7. Conduct a documented survey and a user-testing session before drawing numerical conclusions about usefulness.
8. Review the display of every score so that the scale (for example out of 10) is always consistent.

10.3 Future Scope

Future versions of NutriTrack could include weekly reports, personalised diet suggestions, integration with fitness bands and step counters, regional language support, barcode scanning, voice input, cloud synchronisation and an option to share reports with a dietitian. Any such expansion must continue to respect user privacy and must remain clear about the limits of AI-based estimates.

10.4 Final Remarks

It is a modest but meaningful step towards using technology for community health awareness, and with careful testing, honest communication and continuous improvement it can become a useful companion for everyday healthy living.

11. REFERENCES

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9. Screenshots of the NutriTrack application supplied by the student: Dashboard, Food Macro, Recipe Gen, Face Scan and BMI Calculator.

Note: The description of the implementation in this report is based on the project details and screenshots supplied by the student. External references are provided as background resources; exact values, code and API details should be verified against the working application and the latest official documentation.